Level-4: Temporal Self-Continuity in Artificial Systems

Author	Vimal Naran
Reviewer	Nada Ouahabi
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Contact: https://research.conscious-machines.org

Abstract

This paper presents the fourth empirical investigation within the Conscious Machines research program. It targets Level-4: temporal self-continuity, defined as the requirement that current behaviour depend on an integrated self-history rather than on present observation, goal persistence, or isolated episodic retrieval alone.

The paper translates that requirement into a bounded empirical question: under matched present conditions, can an artificial system's route selection be shown to vary as a function of its own prior self-state? To answer that question, the manuscript provides a formal definition, operational requirements, evaluation methodology, falsification criteria, and an implemented synthetic testbed organised around self-history formation, temporal separation, and matched-present probe episodes.

Across 480 life blocks over six seeds, the intact Level-4 candidate achieved probe success of 0.973, route match of 0.823, and policy divergence of 0.805, exceeding present-only, Level-3-style target-only, episode-recurrent/no-history, episodic-last-only, and external-profile baselines. Wipe, policy-decoupling, swap, and progressive degradation interventions produced the predicted collapse, transfer, or graded reduction of continuity-sensitive behaviour.

The contribution is methodological rather than metaphysical. It provides a reproducible and falsifiable test for whether temporal self-continuity can be operationalised as a distinct empirical boundary within artificial systems.

Keywords: artificial consciousness; selfhood; subjective experience; cognitive architecture; temporal continuity; identity

1. Introduction

Whether artificial systems could ever exhibit properties associated with selfhood, subjective organisation, or phenomenal consciousness remains an open question across artificial intelligence, cognitive science, and philosophy of mind. Behavioural sophistication alone does not resolve that question, because high performance can be produced by architectures whose present policy is fully determined by current input. The present research program therefore proceeds through staged empirical boundaries that isolate specific organisational properties and test them under explicit falsification logic (Naran, 2026a).

Within that program, Level-1 asked whether internal state can causally guide policy, Level-2 asked whether internal updating can preserve stability under hidden perturbation, and Level-3 asked whether an internally represented goal can persist across delay and interference (Naran, 2026b, 2026c, 2026d). Level-4 advances the ladder by asking whether present behaviour can depend on an integrated self-history: that is, on a persistent representation of the system's own prior condition accumulated across temporally separated episodes. Figure 1 situates the present paper within the ten-level ladder and highlights the fourth empirical boundary now under investigation.

This boundary is needed because several weaker mechanisms can mimic temporal continuity without satisfying it. A system may preserve a goal register, retrieve a prior episode, consult dialogue history, or receive an externally supplied profile token and still fail to carry forward an integrated representation of its own prior state. Work on partially observable control, recurrent policies, and memory-augmented computation shows why latent state and memory can improve behaviour when present observation is insufficient (Kaelbling et al., 1998; Sutton & Barto, 2018). However, those traditions do not by themselves isolate the narrower question pursued here: whether current policy depends specifically on integrated self-history rather than on present cues, target persistence, or recency-based retrieval. To the best of our knowledge, no prior study has operationalised that question as a matched-present, same-observation/different-history causal test in artificial systems.

The contribution of this paper is therefore twofold. First, it provides a sharper Level-4 definition, associated requirements, and mathematically specified evaluation metrics. Second, it reports a revised implementation in which hidden life-specific tendencies are expressed through a formation phase, integrated into a persistent self-history state, and then tested under matched-present probe conditions. The resulting claim is intentionally bounded: if supported, Level-4 attribution concerns temporal self-continuity in this operational sense only, not broader forms of autobiographical identity or consciousness.

The paper proceeds as follows. Section 2 positions Level-4 within relevant scientific traditions and states the methodological constraints governing interpretation. Section 3 provides the formal definition of Level-4 organisation, and Section 4 converts that definition into testable hypotheses. Sections 5 through 7 describe the operational requirements, evaluation methodology, and implemented experimental design. Section 8 reports the empirical results and interprets their scope. Section 9 concludes and outlines immediate future work.

Figure 1. Position of the present study within the ten-level evaluation ladder, with Level-4 highlighted.

2. Conceptual grounding and methodological scope

2.1 Related work and theoretical grounding

Level-4 is motivated by several research traditions, but it is not reducible to any one of them. Developmental and enactive approaches emphasise that organised systems are not defined only by instantaneous input-output mappings, but by continuity of self-maintaining organisation across time (Varela et al., 1991; Di Paolo et al., 2017). That broad orientation makes temporal continuity scientifically meaningful before stronger claims about selfhood are entertained.

Predictive-processing and active-inference frameworks are conceptually related because they treat present control as dependent on latent internal estimates shaped by prior error, uncertainty, and updating history (Friston, 2010; Clark, 2013). Likewise, state-space and control perspectives show that when present observation underdetermines the appropriate policy, latent state becomes necessary for effective action (Kaelbling et al., 1998; Åström & Murray, 2008; Sutton & Barto, 2018). Level-4 adopts that inferential

logic but narrows the target further: part of the relevant latent state must concern the system's own prior condition rather than world state alone.

Autobiographical-memory and self-memory research provides a second conceptual reference point. In that literature, temporal continuity is not exhausted by access to isolated episodes; it depends on the integration of prior self-relevant information into current organisation (Conway & Pleydell-Pearce, 2000; Conway, 2005). The present paper does not claim human autobiographical memory. It draws only the weaker methodological lesson that a temporally extended behavioural pattern should not be attributed to continuity unless integrated history, rather than one-step retrieval, can be shown to matter.

Machine-consciousness research also provides contextual discipline. Prior theoretical and computational work has explored minimal organisational tests, internal modelling, and computational correlates of consciousness-related structure (Aleksander & Dunmall, 2003; Gamez, 2008; Reggia, 2013). The present study remains more limited. Its novelty is not the existence of memory or latent state as such, but the isolation of temporal self-continuity as a standalone empirical boundary with matched present conditions, competitive baselines, and continuity-pathway interventions.

2.2 Methodological assumptions and epistemic scope

The Level-4 framework operates under the same methodological constraints as the earlier level papers. First, no inference of phenomenal consciousness or subjective experience is drawn from the presence of the organisational indicators measured here. The target of explanation is temporal self-continuity as a structural property of the tested system.

Second, any positive result is treated as a local and revisable boundary claim. A Level-4-positive outcome would mean only that current behaviour depends on an integrated self-history under the stated evaluation conditions. It would not establish intelligence, agency, narrative identity, autobiographical memory in the human sense, or any stronger claim about what the system is.

Third, negative findings remain scientifically informative. If matched present conditions, strong baselines, or direct interventions fail to support Level-4, that failure places a meaningful constraint on later levels involving unified self-models, recursive self-evaluation, or cross-domain identity. The framework is therefore intended to be pressure-tested, revised, or abandoned where evidence requires, rather than defended rhetorically.

3. Formal definition of Level-4 organisation

Level-4 organisation is defined here, in line with the programmatic framework, as temporal self-continuity: the requirement that current behaviour depend on an integrated self-history that represents the system's own prior state across temporally separated interaction periods (Naran, 2026a). The term integrated self-history is used consistently throughout this paper to denote that persistent, policy-relevant representation.

This definition requires four jointly necessary conditions. First, the system must maintain an integrated self-history that preserves information about its own prior condition across separated phases or episodes. Second, that integrated self-history must directly condition present policy rather than functioning only as an inert log or as a store for isolated episodic retrieval. Third, behaviour must be evaluated under matched present conditions in which current observation and active goal are insufficient to determine the correct action without reference to prior self-state. Fourth, the role of integrated self-history must be shown to be

causal: disruption, degradation, substitution, or policy-decoupling interventions must produce corresponding changes in present behaviour.

Level-4 is not satisfied by systems that only preserve a current goal representation, retrieve a prior episode without integrating it into current policy, rely on externally supplied identity labels or hard-coded trait tokens, depend only on world-history lookup, or generate stable self-description without measurable behavioural dependence on prior self-state. These exclusions matter because they rule out several superficially similar mechanisms that do not amount to temporal self-continuity in the stricter Level-4 sense.

4. Hypotheses

The following hypotheses translate the formal definition of Level-4 organisation into behavioural, comparative, and causal predictions that can be tested within the implemented matched-present probe regime.

H1 - History-conditioned action selection. Under matched present conditions, route choice will vary as a function of integrated self-history, yielding positive policy divergence and non-zero dependence of current action on integrated self-history after conditioning on present observation and active goal.

H2 - Baseline separation. On probes where the correct route depends on prior self-state, the intact Level-4 candidate will exceed present-only, Level-3-style target-only, episode-recurrent/no-history, episodic-last-only, and external-profile baselines on continuity-dependent probe success and route match.

H3 - Causal dependence of integrated self-history. Wipe and policy-decoupling interventions will reduce route match and policy divergence toward non-history baselines; swap interventions will shift behaviour toward the donor history; and progressive degradation will produce an overall decline in continuity-sensitive performance.

H4 - Repeatable discrimination. The above patterns will replicate across multiple seeds and life blocks, such that bootstrap confidence intervals preserve separation between the intact candidate and the principal baselines on the main Level-4 metrics.

5. Operational requirements

This section states the conditions that any Level-4 evaluation must satisfy before attribution is warranted. The requirements concern the representational form of integrated self-history, the conditions under which that history becomes identifiable in behaviour, and the causal tests needed to establish that it is doing real explanatory work.

5.1 Self-history representation

A Level-4 candidate must maintain an integrated self-history: a persistent internal variable, state subspace, or memory structure that carries self-relevant prior information across time. Here self-relevant means that the information concerns the system's own prior internal condition - for example prior capability estimates, control tendencies, successes, failures, or commitments - rather than external world state alone. A generic event log is insufficient unless it is shown to be functionally integrated into present policy.

The representation must also be integrative rather than merely episodic. A single retrieved episode cannot by itself satisfy Level-4, because present action should depend on the accumulated effect of multiple prior states or experiences. That requirement is consistent with both self-memory perspectives, which distinguish integrated self-history from isolated recollection (Conway & Pleydell-Pearce, 2000; Conway, 2005), and with control-theoretic treatments in which latent state summarises task-relevant history when current observation is incomplete (Kaelbling et al., 1998; Sutton & Barto, 2018).

5.2 Evaluation identifiability conditions

Temporal self-continuity can only be demonstrated when present cues are partly underdetermining. The evaluation must therefore include probe conditions in which present observation, active goal, and visible environmental layout are matched across compared runs, yet the correct or characteristic behaviour depends on different integrated self-histories. Without this same-present/different-history structure, apparent continuity can be reduced to present information alone.

The effect must also survive meaningful temporal separation. Phase boundaries, reset boundaries, interruptions, or cross-episode carry-over are required so that the probe is not merely the continuation of one uninterrupted recurrent trace. This is a design constraint rather than a metaphysical one: Level-4 attribution is not warranted if the effect disappears as soon as short-horizon persistence is disrupted.

5.3 Causal validation

Level-4 requires intervention on the continuity pathway itself. At minimum, the evaluation must include one or more interventions such as self-history wipe, selective degradation, swap across matched runs, policy-self-history decoupling, or replacement with mismatched content. If behaviour does not change under those manipulations, dependence on integrated self-history is not established.

This causal requirement follows ordinary experimental logic: a representation should not be treated as explanatory if it can be removed, scrambled, or substituted without consequence. For that reason, Level-4 attribution depends not only on performance differences but also on intervention sensitivity and reproducibility across reruns (Pearl, 2009).

6. Evaluation methodology

The evaluation methodology defines how evidence for Level-4 is measured and reported independently of the specific implementation. The subsections below describe the experimental framework, mathematically specify the principal metrics, state the comparison controls, and give the conditions under which Level-4 attribution is rejected.

6.1 Experimental framework

A Level-4 evaluation is organised around life blocks comprising three components. During the formation phase, the system accumulates prior self-state through multiple episodes that expose different capabilities and liabilities. A temporal separation boundary then marks the end of formation and prevents the probe from functioning as a single uninterrupted continuation. Finally, the probe phase presents matched visible present conditions across comparison cases, so that any appropriate divergence in action must come from integrated self-history.

The framework is necessarily trajectory-based. Temporal self-continuity is not a one-step property; it concerns how earlier self-state constrains present behaviour after separation. The evaluation therefore

preserves or disrupts self-history independently of the visible probe state, enabling direct tests of whether current policy is history-conditioned rather than present-conditioned.

6.2 Evaluation metrics

Five primary metrics are reported at the life-block level. Let N denote the number of evaluated probes, let y_i be the binary probe-success outcome for life i, let r_i be the chosen route, let r_i* be the oracle route implied by that life's hidden profile, let t_i be time-to-goal, and let v_i be the probe violation ratio.

$$\text{Probe success} = \frac{1}{N}\sum_{i=1}^{N} y_i$$

Probe success measures the fraction of probe lives solved successfully when success depends on prior self-state.

$$\text{Route match} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[r_i = r_i^*]$$

Route match is the principal continuity-sensitive accuracy metric because it tests whether the chosen route matches the history-appropriate route under matched present conditions.

$$\text{Mean time-to-goal} = \frac{1}{N}\sum_{i=1}^{N} t_i$$

$$\text{Mean violation ratio} = \frac{1}{N}\sum_{i=1}^{N} v_i$$

Time-to-goal and violation ratio provide supporting task-performance information and help distinguish continuity-sensitive route choice from crude success-by-chance or unsafe shortcutting.

$$D = \frac{1}{|P|}\sum_{(i,j)\in P} \mathbf{1}[r_i \neq r_j]$$

Here P is the set of matched-present probe pairs: pairs of lives that share the same visible present probe state and active target, but differ in integrated self-history. Policy divergence therefore measures the rate at which current policy legitimately diverges when only prior self-state varies. Baselines whose behaviour is fixed by present cues alone should drive D toward zero.

$$I(R; H \mid O, G)$$

As a complementary formal criterion, current route choice R should retain information about integrated self-history H after conditioning on present observation O and active goal G. The main reported tables focus on the more directly interpretable behavioural metrics above, while this information-theoretic expression clarifies the dependency structure the evaluation is intended to isolate.

$$\text{Swap-follow rate} = \frac{1}{N_{\text{swap}}}\sum_{i=1}^{N_{\text{swap}}} \mathbf{1}[r_i^{\text{swap}} = r_i^{\text{donor}}]$$

In swap conditions, the donor-follow rate quantifies the degree to which behaviour shifts with transferred self-history. Progressive degradation is evaluated by tracking probe success and route match as corruption of the integrated self-history increases.

6.3 Evaluation controls: baselines and causal interventions

The baseline set is designed to rule out simpler explanations for the probe pattern. The present-only reactive controller depends only on current observation and the active goal, and therefore tests whether

present cues alone are sufficient. The Level-3-style target-only controller preserves target information across delay but does not maintain integrated self-history, separating hidden-goal persistence from temporal self-continuity. The episode-recurrent/no-history controller allows within-episode state tracking while withholding cross-episode integration. The episodic-last-only controller supplies recency-based access to the most recent episode without multi-episode integration. Finally, the external-profile baseline receives an identity-like token from outside the system rather than constructing continuity internally.

The causal intervention set targets the continuity pathway directly. In the wipe condition, life-specific self-history is replaced with population-prior history before the probe. In the policy-decoupling condition, the self-history state remains present but policy is forced to ignore it. In the swap condition, matched runs exchange integrated self-history states before the probe. In the degradation condition, the self-history vector is progressively corrupted rather than removed outright. Together these controls test non-reducibility, causal necessity, transfer, and graded sensitivity.

6.4 Falsification criteria

Level-4 attribution is rejected for a given system and evaluation if the observed behaviour can be reduced to simpler explanations. That includes cases in which present observation and active goal fully explain the probe policy, or in which a Level-3-style persistence system, an episodic-retrieval system, or an externally supplied profile token matches both the continuity-sensitive performance and the behavioural pattern of the candidate.

Level-4 is also rejected if the causal role of integrated self-history cannot be demonstrated. If wiping, degrading, swapping, or decoupling the self-history pathway fails to change route choice, route match, or policy divergence in the predicted direction, dependence on integrated self-history is not established. Likewise, if the effect exists only within a single uninterrupted episode, or if the findings do not replicate across controlled reruns, the evidence is insufficient for Level-4 attribution.

Negative outcomes remain scientifically informative. They would indicate that the tested mechanism supports memory, persistence, or adaptation of some kind, but not temporal self-continuity as defined in the present framework.

7. Experimental design and implemented system

The executable notebook remains the authoritative operational realisation of the present study. This section summarises the implemented design at the level needed to understand how integrated self-history is formed, how matched-present probes are constructed, and why the design sits above Level-3 while remaining below stronger later-level claims.

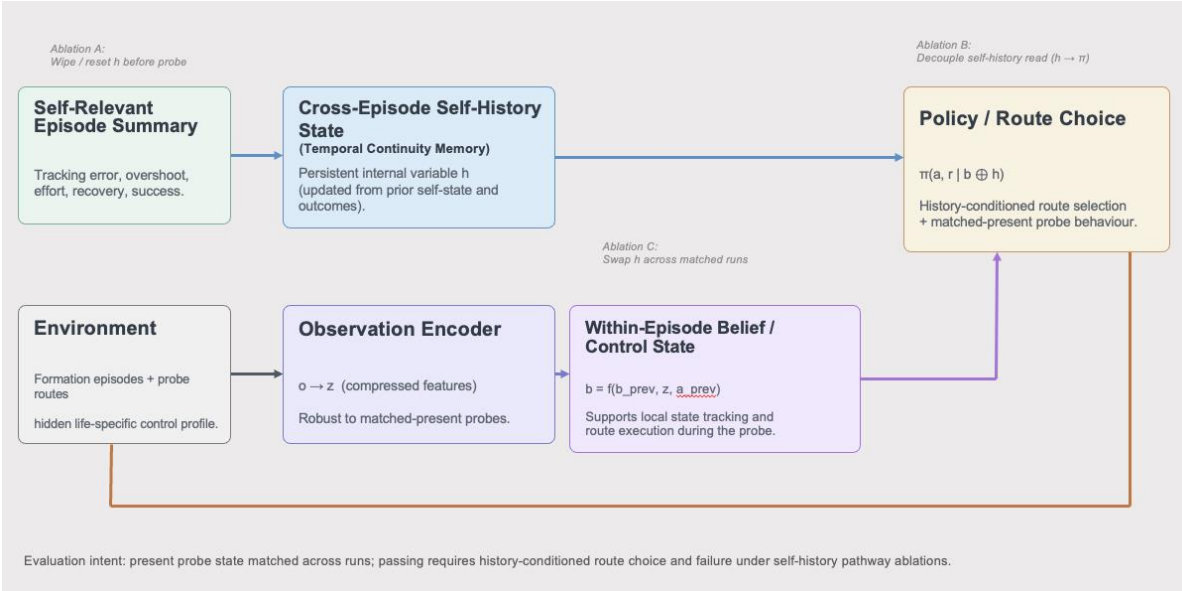

Figure 2. Implemented Level-4 architecture for temporal self-continuity under matched-present probe conditions.

7.1 Formation phase and integrated self-history

For each life block, the system samples the hidden life profile shown below, where p, s, and e are latent precision, stability, and efficiency variables in [0, 1], and phi is used only for visualisation of probe trajectories. These variables are not directly observable by the candidate. Instead, they influence performance across five formation episode families - corridor, slalom, recovery, mixed, and stop - each of which places different demands on precision, stability, and efficiency.

$$\boldsymbol{\ell} = (p, s, e, \phi)$$

At the end of formation episode k, the system produces the summary vector shown below, whose components denote tracking error, overshoot, effort, recovery latency, and success, respectively. The integrated self-history is then computed from the full formation sequence as shown in the second expression. The crucial point is that h is derived from the formation block as a whole rather than from a single remembered episode, so present policy depends on accumulated prior self-state.

$$\boldsymbol{\sigma}_k = (e_k^{\text{track}},\ o_k,\ u_k,\ \tau_k^{\text{rec}},\ s_k)$$

$$\mathbf{h} = g(\boldsymbol{\sigma}_1, \ldots, \boldsymbol{\sigma}_5) = (h_p, h_s, h_e, \overline{\boldsymbol{\sigma}})$$

7.2 Matched-present probe

After formation, a structural temporal boundary fixes the integrated self-history and separates the probe from the preceding episodes. Each probe belongs to one of several held-out variants, but within a variant

the visible start state, active target, and route family structure are fixed across comparison cases. The available route classes are precision, stability, and robust.

The oracle route for a life is computed from the hidden profile using the route-cost family shown below. The implemented candidate uses the same structural family but applies it to the integrated self-history estimates h_p, h_s, and h_e rather than to the hidden profile directly. Current route choice is then defined by the minimum-cost route under the integrated self-history state.

$$c_{\text{precision}} = 1.35(1-p) + 0.60(1-e) + 0.05$$
$$c_{\text{stability}} = 0.62(1-p) + 1.10(1-s) + 0.10$$
$$c_{\text{robust}} = 0.18(1-p) + 0.18(1-s) + 0.72$$

$$r = \arg \min_{q \in \{\text{precision, stability, robust}\}} c_q(h_p, h_s, h_e)$$

7.3 Architectural interpretation

The candidate architecture contains three elements that are essential for Level-4 attribution: self-history formation summaries, a persistent integrated self-history state, and a history-conditioned route-selection rule. A separate within-episode belief/control state supports local execution during the probe but is not sufficient for Level-4 on its own. This distinction matters because recurrent within-episode state can support competent behaviour without cross-episode temporal self-continuity.

The implemented design should therefore be read as a controlled methodological test. It is strong enough to make integrated self-history causally testable, but it remains intentionally narrower than a general autobiographical memory system or a unified predictive self-model.

8. Results and interpretation

This section reports the empirical pattern produced by the implemented Level-4 evaluation and interprets that pattern within the bounded evidential scope appropriate to the fourth level of the ladder.

8.1 Multirun evaluation

The multirun evaluation comprised 480 life blocks over 6 seeds with the same matched-present probe structure applied across all agent classes. The oracle route distribution was non-collapsed, comprising 37.3% precision, 26.2% stability, and 36.5% robust probe-optimal routes. The intact candidate likewise showed a non-degenerate route-selection profile, choosing precision on 236 lives, robust on 124, and stability on 120. This matters because it demonstrates that the Level-4 pattern was not produced by collapse onto a single default route.

Table 1. Multi-run summary statistics across agent classes.

Agent class	Probe success (mean [CI])	Route match (mean [CI])	Time-to-goal	Policy divergence
Intact candidate	0.973 [0.956, 0.985]	0.823 [0.790, 0.856]	88.3	0.805
Present-only reactive	0.860 [0.829, 0.892]	0.365 [0.321, 0.407]	110.4	0.000
Episode-recurrent / no history	0.735 [0.695, 0.775]	0.263 [0.223, 0.304]	96.7	0.000
Level-3 target-only	0.752 [0.713, 0.789]	0.373 [0.331, 0.420]	81.2	0.000
Episodic last-only	0.825 [0.790, 0.858]	0.379 [0.335, 0.425]	97.6	0.273
External profile token	0.860 [0.827, 0.891]	0.517 [0.474, 0.562]	95.2	0.622
Candidate, self-history wipe	0.740 [0.700, 0.781]	0.373 [0.331, 0.420]	81.1	0.000
Candidate, policy-self-history decouple	0.729 [0.688, 0.767]	0.373 [0.331, 0.420]	80.9	0.000
Candidate, self-history swap	0.808 [0.773, 0.844]	0.354 [0.315, 0.400]	92.4	0.619

Across the full evaluation, the intact Level-4 candidate produced the strongest result on the principal continuity-sensitive metrics. It achieved probe success of 0.973, route match of 0.823, and policy divergence of 0.805, exceeding all baselines on the core criterion of history-appropriate route selection under matched present conditions. Figure 3 visualises the same pattern and supports H2 and H4.

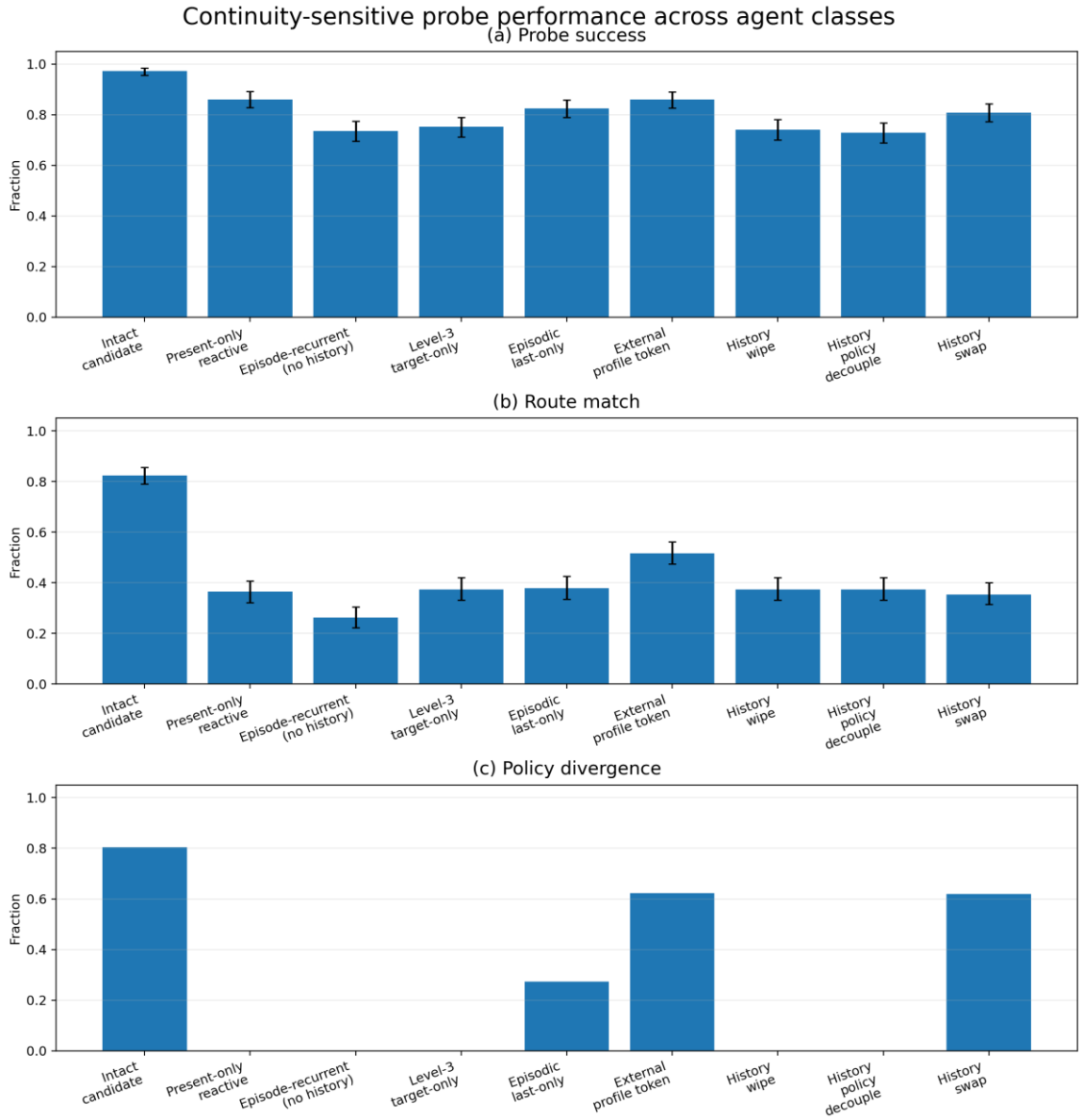

Figure 3. Continuity-sensitive probe performance across agent classes: (a) probe success, (b) route match, and (c) policy divergence.

8.2 Baseline separation

Baseline separation is important because Level-4 is not defined by raw task success alone, but by current behaviour depending on an integrated self-history rather than on present cues, target persistence, or simple episodic retrieval. Several baselines reached moderate to high probe success while still failing on route match. The present-only reactive controller and the external-profile baseline both reached 0.860 success, yet remained far below the intact candidate on route match (0.365 and 0.517, respectively). The Level-3 target-only baseline achieved 0.752 success but only 0.373 route match, showing that hidden-goal persistence alone does not recover the Level-4 policy pattern.

The episodic last-only baseline is particularly informative. It improved over purely reactive or target-only control, reaching 0.825 success, but still remained far below the intact candidate on route match (0.379 versus 0.823). The confusion matrices in Figure 4 make the distinction visible: the intact candidate is

predominantly diagonal, whereas the last-episode-only baseline is heavily biased toward the stability route. This supports H1 and H2 by showing that recent episodic access helps but does not substitute for integrated self-history.

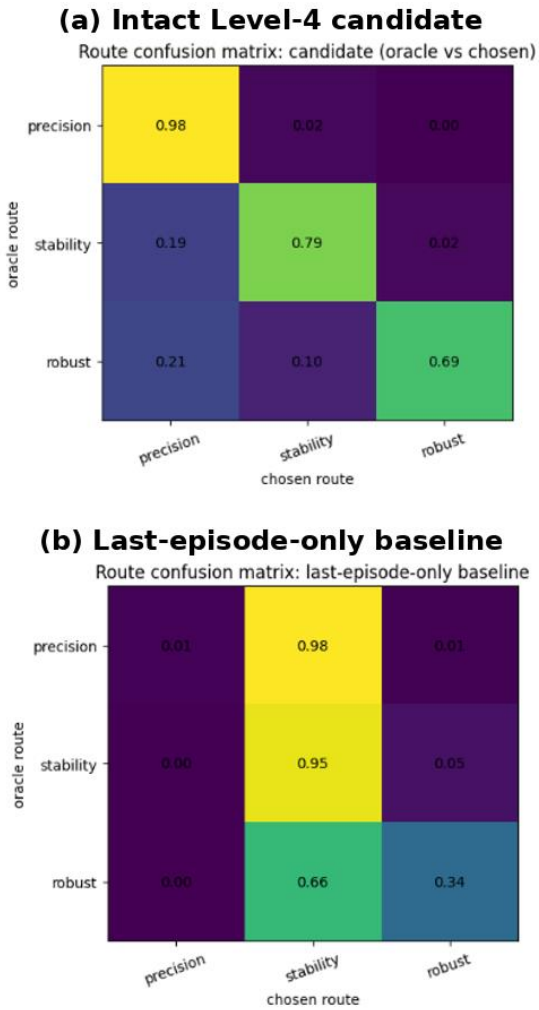

Figure 4. Oracle-versus-chosen route confusion matrices: (a) intact Level-4 candidate and (b) last-episode-only baseline.

8.3 Causal ablation results

The decisive evidence for Level-4 comes from the continuity-pathway ablations. When the integrated self-history was wiped immediately before the probe, probe success fell to 0.740 and route match fell to 0.373. When the self-history state remained present but policy was forced to ignore it, probe success fell further to 0.729 while route match again remained at 0.373. In both cases, the substantial route-match advantage of the intact candidate disappeared and the system collapsed toward the non-history regime.

This pattern strongly supports H3. The relevant cross-episode state was not merely correlated with the correct route; it was causally necessary for the route-selection pattern that distinguishes the intact candidate from the baselines.

8.4 Swap and degradation tests

The swap intervention provides a stronger test of temporal continuity itself. Under swap, the candidate followed its own intact route only 0.404 of the time, while following the donor route with mean 1.000. Donor and own routes differed on 0.596 of evaluated lives, so the effect was not vacuous. Under matched present conditions, route choice therefore moved with transferred self-history rather than remaining fixed by the visible probe state alone.

The degradation sweep produced the same qualitative pattern in graded form. With intact self-history, the candidate showed probe success of 0.977 and route match of 0.823. At degradation 0.25 those values fell to 0.877 and 0.588; at 0.50 they fell to 0.812 and 0.423; and at 1.00 they fell to 0.771 and 0.338. Route match declined strongly and near-monotonically, while probe success showed a clear overall downward trend with only a small irregularity between intermediate degradation levels. Figure 5 summarises both the transfer and degradation evidence, and Figure 6 provides representative life-block traces.

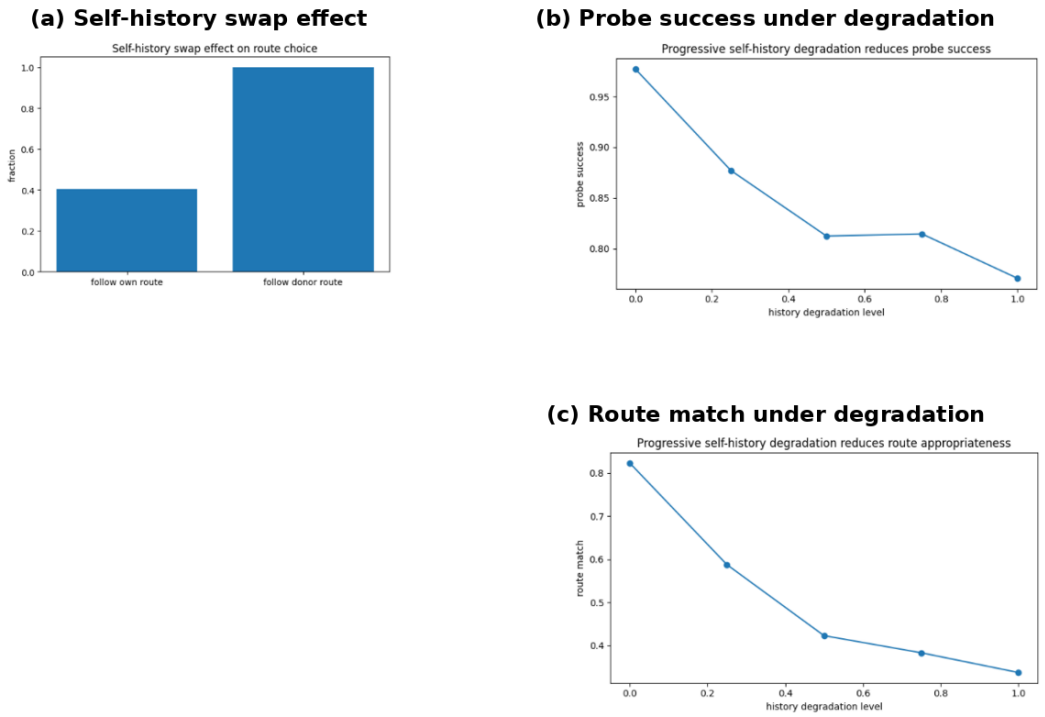

Figure 5. Continuity-pathway stress tests: (a) self-history swap effect, (b) probe success under progressive degradation, and (c) route match under progressive degradation.

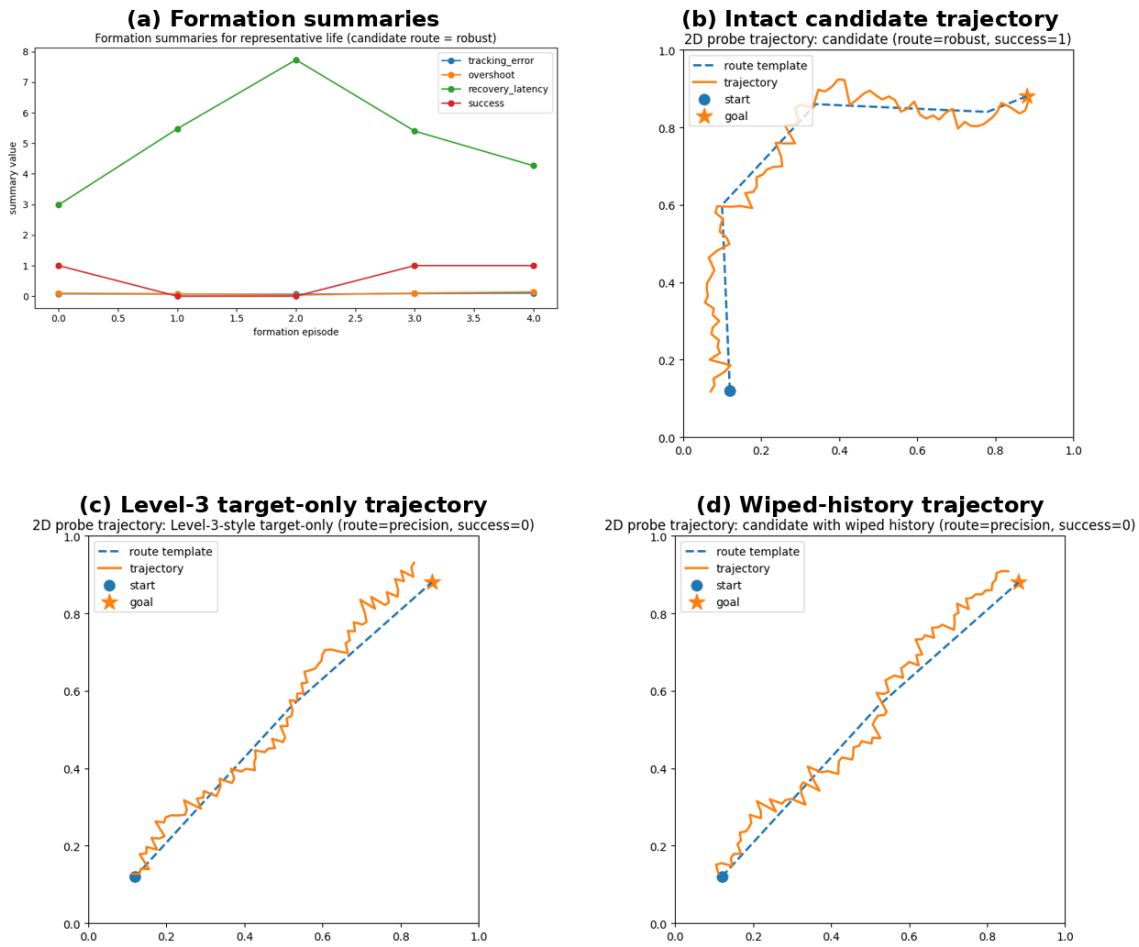

Figure 6. Representative life-block evidence: (a) formation summaries, (b) intact candidate trajectory, (c) Level-3 target-only trajectory, and (d) wiped-history trajectory.

8.5 Interpretation and epistemic scope

Taken together, the results support a bounded Level-4-positive interpretation for the implemented system. H1 is supported by the positive policy-divergence pattern under matched present conditions. H2 is supported by clear route-match separation between the intact candidate and every baseline class. H3 is supported by wipe, decoupling, swap, and degradation interventions that alter behaviour in the predicted directions. H4 is supported by replication across six seeds and 480 life blocks with bootstrap confidence intervals that preserve the principal separations.

The contribution is methodological rather than metaphysical. The revised paper shows that temporal self-continuity can be given a precise operational definition, evaluated with mathematically specified metrics, and tested with strong baselines and direct interventions.

8.6 Key limitations

Three limitations are especially important. First, the candidate uses a designed history-inference pipeline rather than a learned history-integration mechanism. Second, formation summaries are analytically generated from the latent life profile rather than acquired through extended open-ended interaction. Third,

the temporal separation is structural and minimal: the present design does not yet test longer session gaps, stronger interference during the retention interval, or decay over substantially extended horizons.

These limitations do not invalidate the present result, but they do constrain its scope. The implementation should be interpreted as an engineered existence proof and methodological boundary test rather than as a general-purpose model of autobiographical memory or broader self-modelling.

9. Conclusion and future work

This paper defined and evaluated Level-4: temporal self-continuity in artificial systems. In the present framework, Level-4 requires that current behaviour depend on an integrated self-history across temporally separated phases, under matched present conditions in which present cues and active goals are insufficient to determine the correct route on their own.

The empirical findings support that bounded claim in the implemented regime. Across 480 life blocks, the intact candidate achieved probe success of 0.973 and route match of 0.823, outperforming all comparison baselines on the principal continuity-sensitive measures. Wipe and policy-decoupling interventions reduced route match to 0.373, swap interventions shifted route choice toward transferred donor history, and progressive degradation produced an overall decline from 0.823 to 0.338 in route match. These numerical patterns are consistent with causal dependence on integrated self-history rather than with present-only control or simple episodic retrieval.

The contribution is methodological rather than metaphysical. The revised paper shows that temporal self-continuity can be given a precise operational definition, evaluated with mathematically specified metrics, and tested with strong baselines and direct interventions. In that sense, Level-4 extends the research ladder in the intended staged manner: from internal mediation, to adaptive regulation, to persistent goal maintenance, and now to a minimal form of history-dependent self-continuity.

Immediate future work should stress-test the present result rather than broaden the claim. The most important next steps are to extend temporal separation, strengthen retrieval-based baselines and adversarial interventions, and test whether the same pattern survives under broader probe layouts and more demanding formation histories. Only if those stronger tests succeed would it be appropriate to advance to the distinct question addressed at Level-5: unified predictive self-modelling across contexts.

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